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Bitcoin Price Prediction Using Machine Learning Models and Stochastic Differential Equations

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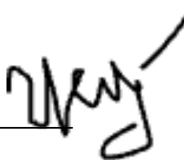
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Abstract

This thesis investigates the problem of Bitcoin price prediction through a dual lens: classical stochastic differential equation (SDE) models drawn from mathematical finance, and modern machine learning (ML) methods. The central theoretical contribution is a rigorous justification of why cryptocurrency prices should be modelled as stochastic processes: we review the Efficient Market Hypothesis, the Random Walk Hypothesis, and the empirical stylised facts of Bitcoin log-returns - heavy tails, volatility clustering, and sudden price jumps - and argue that each of these phenomena naturally motivates a specific class of continuous-time stochastic models.

Three SDE models are studied in detail: Geometric Brownian Motion (GBM), the Merton Jump-Diffusion model, and the Heston Stochastic Volatility model. For each model we state the driving assumptions as formal propositions, derive the governing equation, present the calibration procedure, and prove the key existence and uniqueness results for the associated SDE.

Four ML models - Elastic Net, Random Forest, Gradient Boosting, and Long Short-Term Memory (LSTM) networks - are trained on multi-resolution Bitcoin price data obtained from Yahoo Finance and evaluated alongside the SDE models using rolling one-step-ahead forecasts. Comparative analysis on the metrics MAE, RMSE, and R^2 shows that ML models achieve superior point-prediction accuracy on short horizons, while SDE models provide distributional forecasts and interpretable economic parameters.

Keywords: Bitcoin, cryptocurrency, stochastic differential equations, Geometric Brownian Motion, Merton model, Heston model, machine learning, LSTM, time series forecasting, random walk.

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Introduction

Motivation

Bitcoin was introduced in 2008 by the pseudonymous Satoshi Nakamoto as the first decentralised digital currency [1]. In the sixteen years since its creation, Bitcoin has grown from an obscure cryptographic experiment into a global asset class with a market capitalisation exceeding one trillion US dollars. Its price exhibits a unique combination of extreme volatility, structural breaks, speculative bubbles, and rapid recoveries that challenge traditional financial modelling paradigms.

The question of whether Bitcoin prices can be predicted - and, if so, by what class of models - has attracted considerable academic interest. From the perspective of mathematical finance, the natural starting point is to ask: *Is Bitcoin price a stochastic process, and if so, which one?* Answering this question rigorously requires both theoretical arguments (grounded in the theory of financial markets) and empirical evidence (derived from the statistical properties of observed price data).

Problem Statement

Let S_t denote the Bitcoin price denominated in US dollars at time t . The fundamental modelling problem may be stated as follows.

1. **(Theoretical)** Under which assumptions on market structure and investor behaviour does $\{S_t\}_{t \geq 0}$ admit a representation as the solution of a stochastic differential equation?
2. **(Statistical)** Do the empirical log-returns of Bitcoin, $r_t = \ln(S_t/S_{t-1})$, satisfy the probabilistic conditions implied by candidate SDE models (e.g. independent increments, Gaussianity, stationarity)?
3. **(Predictive)** Among the class of SDE models and data-driven ML models, which achieves the best forecasting performance on a held-out test set, as measured by standard regression metrics?

Research Objectives

The specific objectives of this thesis are:

- to provide a self-contained mathematical introduction to stochastic processes, Itô calculus, and SDEs at the level required for the models under study;

- to rigorously justify, via explicit propositions and proofs, the modelling assumption that cryptocurrency prices follow stochastic processes;
- to derive, calibrate, and implement three SDE models: GBM, Merton Jump-Diffusion, and the Heston model;
- to build, train, and evaluate four ML models on real Bitcoin price data;
- to compare SDE and ML approaches on a common evaluation framework, identifying the strengths and limitations of each paradigm.

Structure of the Thesis

Chapter 1 develops the mathematical foundations: probability spaces, filtrations, Brownian motion, stochastic integration, and Itô's lemma. Chapter 2 presents the core theoretical argument for modelling Bitcoin as a stochastic process, including the Efficient Market Hypothesis, the Random Walk Hypothesis, and a review of empirical stylised facts. Chapter 3 derives and analyses the three SDE models. Chapter 4 describes the four ML models. Chapter 5 covers data, feature engineering, and evaluation protocols. Chapter 6 presents the experimental results. The Conclusion summarises findings and outlines directions for future work.

Chapter 1

Mathematical Foundations of Stochastic Processes

1.1 Probability Spaces and Filtrations

Definition 1.1 (Probability Space). A *probability space* is a triple $(\Omega, \mathcal{F}, \mathbb{P})$ where Ω is a non-empty set of elementary outcomes, $\mathcal{F}$ is a σ -algebra of subsets of Ω (events), and $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ is a probability measure satisfying $\mathbb{P}(\Omega) = 1$ and countable additivity.

Financial modelling requires tracking the evolution of information over time.

Definition 1.2 (Filtration). Let $T > 0$. A *filtration* on $(\Omega, \mathcal{F}, \mathbb{P})$ is a family $\mathbb{F} = (\mathcal{F}_t)_{0 \leq t \leq T}$ of sub- σ -algebras of $\mathcal{F}$ satisfying $\mathcal{F}_s \subseteq \mathcal{F}_t$ for all $s \leq t$. The filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ is said to satisfy the *usual conditions* if $\mathcal{F}_0$ contains all $\mathbb{P}$ -null sets and the filtration is right-continuous: $\mathcal{F}_t = \bigcap_{s>t} \mathcal{F}_s$ for all t .

Definition 1.3 (Stochastic Process). A *stochastic process* on $(\Omega, \mathcal{F}, \mathbb{P})$ is a collection of random variables $\{X_t\}_{t \in \mathcal{T}}$ indexed by a parameter set $\mathcal{T}$ (typically $[0, T]$ or $\mathbb{Z}_{\geq 0}$). The process is said to be $\mathbb{F}$ -*adapted* if X_t is $\mathcal{F}_t$ -measurable for every t .

The concept of adaptedness captures the intuition that at time t , the value X_t is determined by the information available up to that moment.

1.2 Martingales

Definition 1.4 (Martingale). An adapted, integrable process $\{M_t\}$ is a *martingale* with respect to $\mathbb{F}$ if

$$\mathbb{E}[M_t \mid \mathcal{F}_s] = M_s \quad \text{for all } 0 \leq s \leq t.$$

It is a *supermartingale* (resp. *submartingale*) if the equality is replaced by $\leq$ (resp. $\geq$).

The martingale property is intimately related to the notion of a fair game: in a frictionless arbitrage-free market, discounted asset prices are martingales under a risk-neutral probability measure.

This observation, formalised in the First Fundamental Theorem of Asset Pricing, provides one of the deepest theoretical justifications for modelling asset prices as stochastic processes.

1.3 The Wiener Process (Brownian Motion)

The Wiener process is the canonical building block of continuous-time stochastic models in finance.

Definition 1.5 (Standard Wiener Process). A stochastic process $\{W_t\}_{t \geq 0}$ on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ is a *standard Wiener process* (or *standard Brownian motion*) if:

(W1) $W_0 = 0$ almost surely;

(W2) W has *independent increments*: for $0 \leq t_0 < t_1 < \dots < t_n$, the increments $W_{t_1} - W_{t_0}, \dots, W_{t_n} - W_{t_{n-1}}$ are mutually independent;

(W3) W has *stationary Gaussian increments*: for all $0 \leq s < t$,

$$W_t - W_s \sim \mathcal{N}(0, t - s);$$

(W4) W has *continuous paths*: $t \mapsto W_t(\omega)$ is continuous for $\mathbb{P}$ -almost every $\omega \in \Omega$.

Proposition 1.6 (Martingale Property of Brownian Motion). A standard Wiener process $\{W_t\}$ is a martingale with respect to its natural filtration $\mathcal{F}_t^W = \sigma(W_s : 0 \leq s \leq t)$.

Proof. For $s \leq t$, write $W_t = W_s + (W_t - W_s)$. By (W2), $(W_t - W_s)$ is independent of $\mathcal{F}_s^W$. By (W3), $\mathbb{E}[W_t - W_s] = 0$. Therefore,

$$\mathbb{E}[W_t \mid \mathcal{F}_s^W] = W_s + \mathbb{E}[W_t - W_s \mid \mathcal{F}_s^W] = W_s + \mathbb{E}[W_t - W_s] = W_s. \quad \square$$

Proposition 1.7 (Quadratic Variation of Brownian Motion). For a standard Wiener process, the quadratic variation over $[0, t]$ satisfies

$$\langle W \rangle_t := \lim_{\|\Pi\| \rightarrow 0} \sum_{i=1}^n (W_{t_i} - W_{t_{i-1}})^2 = t \quad \text{in } L^2(\mathbb{P}),$$

where $\Pi = \{0 = t_0 < t_1 < \dots < t_n = t\}$ and $\|\Pi\| = \max_i (t_i - t_{i-1})$.

This non-trivial result - that Brownian paths accumulate quadratic variation at the constant rate of one unit per unit of time - is the foundation of Itô's calculus. Informally, the notation $dW_t^2 = dt$ encodes this fact.

Remark 1.8. Brownian motion paths are continuous but nowhere differentiable with probability one. Consequently, the classical Lebesgue-Stieltjes integral $\int_0^t f(s) dW_s$ is not well-defined pathwise, and a new integration theory - the Itô integral - is required.

1.4 The Itô Stochastic Integral

Definition 1.9 (Itô Integral for Simple Processes). Let $\phi = \sum_{i=0}^{n-1} \phi_i \mathbb{1}_{[t_i, t_{i+1})}$ be an $\mathbb{F}$ -adapted step process with $\mathbb{E}[\phi_i^2] < \infty$. The *Itô integral* of ϕ with respect to W is defined as

$$I_t(\phi) := \int_0^t \phi_s dW_s = \sum_{i: t_i < t} \phi_i (W_{t_{i+1} \wedge t} - W_{t_i}).$$

For general adapted square-integrable processes ϕ , the integral is extended by a density argument in the L^2 sense. The key isometry property is stated below.

Proposition 1.10 (Itô Isometry). *For any $\mathbb{F}$ -adapted, square-integrable process ϕ ,*

$$\mathbb{E} \left[\left(\int_0^t \phi_s dW_s \right)^2 \right] = \int_0^t \mathbb{E}[\phi_s^2] ds.$$

This identity implies that $\{I_t(\phi)\}_{t \geq 0}$ is a martingale, and underpins the L^2 theory of stochastic integration.

1.5 Itô's Lemma

Itô's lemma is the stochastic analogue of the chain rule and is the single most important computational tool in continuous-time finance.

Theorem 1.11 (Itô's Lemma - One-Dimensional Form). *Let X_t satisfy the stochastic differential equation $dX_t = \mu_t dt + \sigma_t dW_t$, and let $f : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ be a function of class $C^{1,2}$ (once continuously differentiable in t , twice in x). Then*

$$df(t, X_t) = \frac{\partial f}{\partial t} dt + \frac{\partial f}{\partial x} dX_t + \frac{1}{2} \frac{\partial^2 f}{\partial x^2} \sigma_t^2 dt. \quad (1.1)$$

The extra term $\frac{1}{2}(\partial^2 f / \partial x^2) \sigma_t^2 dt$ arises precisely because of Proposition 1.7: when we expand df using Taylor's theorem, the term $(\partial^2 f / \partial x^2)(dX_t)^2 / 2$ contributes $\frac{1}{2}(\partial^2 f / \partial x^2) \sigma_t^2 dt$ instead of vanishing as it would in classical calculus.

1.6 Existence and Uniqueness for SDEs

Definition 1.12 (Stochastic Differential Equation). A *stochastic differential equation* (SDE) on $[0, T]$ is of the form

$$dX_t = b(t, X_t) dt + \sigma(t, X_t) dW_t, \quad X_0 = x_0 \in \mathbb{R}, \quad (1.2)$$

where $b, \sigma : [0, T] \times \mathbb{R} \rightarrow \mathbb{R}$ are measurable coefficient functions.

Theorem 1.13 (Existence and Uniqueness - Lipschitz Coefficients). *Suppose the coefficients b and σ in (1.2) satisfy:*

(C1) **Global Lipschitz condition:** *there exists $L > 0$ such that*

$$|b(t, x) - b(t, y)| + |\sigma(t, x) - \sigma(t, y)| \leq L|x - y|$$

for all $t \in [0, T]$ and $x, y \in \mathbb{R}$;

(C2) **Linear growth condition:** *there exists $C > 0$ such that*

$$|b(t, x)|^2 + |\sigma(t, x)|^2 \leq C^2(1 + x^2)$$

for all $t \in [0, T]$ and $x \in \mathbb{R}$.

Then equation (1.2) has a unique strong solution $\{X_t\}_{t \in [0, T]}$ that is $\mathbb{F}$ -adapted and satisfies $\mathbb{E}[\sup_{0 \leq t \leq T} X_t^2] < \infty$.

We will verify conditions (C1)–(C2) for each of the three SDE models studied in Chapter 3.

Chapter 2

Why Bitcoin Price Is a Stochastic Process

This chapter presents the theoretical and empirical justifications for treating the Bitcoin price S_t as the solution of a stochastic differential equation. The argument proceeds in three stages.

1. **Market microstructure.** Under the Efficient Market Hypothesis, asset prices incorporate all available information instantaneously, implying that price changes are driven by new, unpredictable information arrivals - naturally modelled as random shocks.
2. **Random Walk Hypothesis.** The discrete-time log-return series $\{r_t\}$ has the statistical properties of an i.i.d. process, at least to a first approximation.
3. **Stylised facts.** The empirical distribution of Bitcoin log-returns exhibits heavy tails, volatility clustering, and occasional large jumps - properties that motivate extensions of the basic Gaussian random walk.

2.1 The Efficient Market Hypothesis

Definition 2.1 (Informational Efficiency [2]). A financial market is said to be *efficient* with respect to an information set Φ_t if asset prices fully and instantaneously reflect all information contained in Φ_t .

Three nested forms are commonly distinguished.

- **Weak-form efficiency:** Φ_t contains only past prices and trading volumes.
- **Semi-strong efficiency:** Φ_t includes all publicly available information (financial statements, macroeconomic data, news).
- **Strong-form efficiency:** Φ_t includes all information, public and private.

Proposition 2.2 (Stochastic Prices under the EMH). *Suppose the market for Bitcoin satisfies weak-form efficiency. Then the price process $\{S_t\}$ is a Markov process, and the log-return process $\{r_t = \ln(S_t/S_{t-1})\}$ constitutes a sequence of random variables uncorrelated with any function of past prices.*

Argument. Under weak-form efficiency, no trading strategy based solely on the history $(S_0, S_1, \dots, S_{t-1})$ can generate expected profits above the equilibrium return. Formally, if π_t denotes the equilibrium expected return per period,

$$\mathbb{E}[S_{t+1} \mid S_0, \dots, S_t] = (1 + \pi_t)S_t.$$

This is the martingale (or supermartingale) property. Since S_{t+1} depends on past prices only through S_t , the process is Markov. Any excess return $r_t - \pi_{t-1}$ is unpredictable from past information, hence random. $\square$

Remark 2.3. Proposition 2.2 does *not* claim that prices are unpredictable in an absolute sense - only that the predictable component is the risk premium. The unpredictable residual, the “news component”, is what stochastic models aim to characterise.

2.2 The Random Walk Hypothesis

In continuous time, the “price driven by random news arrivals” idea leads to the following hypothesis.

Assumption 2.4 (Continuous Random Walk Hypothesis). The log-price process $\{L_t = \ln S_t\}$ satisfies

$$L_t - L_s = \mu(t - s) + \sigma(W_t - W_s), \quad 0 \leq s \leq t,$$

for constants $\mu \in \mathbb{R}$ (drift) and $\sigma > 0$ (diffusion coefficient), where W is a standard Wiener process.

Proposition 2.5 (Equivalence with Geometric Brownian Motion). *Assumption 2.4 is equivalent to the requirement that S_t solves the SDE*

$$dS_t = \mu^* S_t dt + \sigma S_t dW_t, \quad \mu^* = \mu + \frac{1}{2}\sigma^2,$$

i.e., S_t follows Geometric Brownian Motion.

Proof. Apply Itô’s Lemma (Theorem 1.11) to $f(x) = e^x$ with $X_t = \ln S_t = L_t$. From Assumption 2.4, $dL_t = \mu dt + \sigma dW_t$. Then

$$dS_t = d(e^{L_t}) = e^{L_t} dL_t + \frac{1}{2}e^{L_t}(dL_t)^2 = S_t(\mu dt + \sigma dW_t) + \frac{1}{2}S_t\sigma^2 dt = (\mu + \frac{1}{2}\sigma^2)S_t dt + \sigma S_t dW_t. \quad \square$$

2.2.1 Empirical Testing of the Random Walk Hypothesis

The Random Walk Hypothesis (RWH) implies specific testable restrictions on the log-return series. Three classical tests are considered here.

Augmented Dickey-Fuller test. The ADF test examines whether the log-price series contains a unit root. Under the null $H_0: \rho = 1$, the log-price follows a random walk (possibly with drift). Rejection of H_0 in favour of $H_1: \rho < 1$ would indicate mean-reversion, inconsistent with the RWH. For Bitcoin,

empirical studies consistently *fail to reject* H_0 at standard significance levels for prices, while the return series $\{r_t\}$ is stationary.

Ljung-Box test for serial correlation. The Ljung-Box Q -statistic tests the joint hypothesis that the first k autocorrelations of $\{r_t\}$ are zero:

$$Q(k) = n(n+2) \sum_{j=1}^k \frac{\hat{\rho}_j^2}{n-j},$$

where $\hat{\rho}_j$ is the sample autocorrelation at lag j and n is the sample size. Under the RWH, $Q(k) \xrightarrow{d} \chi^2(k)$. Empirical results for Bitcoin show negligible autocorrelation in raw returns, consistent with the RWH, but *significant* autocorrelation in squared returns $\{r_t^2\}$ - the signature of volatility clustering discussed in Section 2.3.

Variance ratio test. If $\{r_t\}$ is i.i.d., the variance of k -period returns equals k times the variance of one-period returns. The variance ratio

$$VR(k) = \frac{\text{Var}[r_t^{(k)}]}{k \cdot \text{Var}[r_t]}, \quad r_t^{(k)} = \sum_{j=0}^{k-1} r_{t-j},$$

should equal one under the RWH. Empirical estimates for daily Bitcoin data typically yield $VR(k) \approx 1$ for moderate k , supporting the RWH in the first-moment sense, although deviations appear at high frequencies.

2.3 Stylised Facts of Bitcoin Log-Returns

While the RWH provides a theoretical justification for stochastic modelling, the Gaussian random walk (Assumption 2.4) is an idealisation that fails to capture several well-documented empirical regularities, known as *stylised facts*.

2.3.1 Heavy Tails (Leptokurtosis)

Proposition 2.6 (Kurtosis of Gaussian Returns). *If $\{r_t\} \stackrel{i.i.d.}{\sim} \mathcal{N}(\mu, \sigma^2)$, then the excess kurtosis is $\kappa = 0$.*

In practice, the excess kurtosis of Bitcoin daily log-returns is $\hat{\kappa} \approx 5\text{--}15$, implying a probability of large moves far in excess of Gaussian predictions. A Jarque-Bera test strongly rejects normality at any conventional significance level.

The heavy-tailed distribution of returns can be attributed to two sources.

- **Jump events:** rare but large discontinuous price movements caused by exchange hacks, regulatory announcements, or macroeconomic shocks. The Merton Jump-Diffusion model (Section 3.2) captures this by superimposing a compound Poisson process on the Brownian diffusion.

- **Stochastic volatility:** if the instantaneous variance v_t is itself random and time-varying, the marginal distribution of returns $r_t = \int_{t-\Delta}^t \sqrt{v_s} dW_s + \mu\Delta$ is a Gaussian scale mixture, which is inherently heavy-tailed. This motivates the Heston model (Section 3.3).

2.3.2 Volatility Clustering

Definition 2.7 (Volatility Clustering). A return series $\{r_t\}$ exhibits *volatility clustering* if large (small) absolute returns tend to be followed by large (small) absolute returns:

$$\text{Corr}(|r_t|, |r_{t-k}|) > 0 \quad \text{for small } k \geq 1.$$

Volatility clustering was first documented by Mandelbrot [3] for commodity prices and has been universally confirmed in cryptocurrency markets. It implies that the instantaneous volatility σ_t is not constant but is itself a stochastic process, providing direct empirical motivation for the Heston model.

Proposition 2.8 (GBM Cannot Exhibit Volatility Clustering). *Under GBM, the log-returns $r_t = (\mu - \frac{1}{2}\sigma^2)\Delta t + \sigma\Delta W$ are i.i.d. Gaussian, so $\text{Corr}(|r_t|, |r_{t-k}|) = 0$ for all $k \geq 1$.*

This proposition shows that GBM is *structurally incapable* of reproducing volatility clustering, and motivates the introduction of stochastic volatility models.

2.3.3 Leverage Effect (Asymmetric Volatility)

Empirical studies of traditional equity markets report a *negative* correlation between return shocks and subsequent volatility changes: large negative returns tend to be followed by elevated volatility. For Bitcoin, this effect is attenuated but still statistically significant, particularly at daily frequencies.

Formally, if $\rho = \text{Corr}(dW_1, dW_2)$ denotes the correlation between the price Brownian W_1 and the variance Brownian W_2 in the Heston model, then $\rho < 0$ captures the leverage effect.

2.3.4 Non-Stationarity and Structural Breaks

Bitcoin prices have undergone several distinct regimes, including explosive growth phases (2017, 2020-2021) and sustained drawdowns (2018, 2022). The log-price series is non-stationary ($I(1)$), while the log-return series is weakly stationary. Structural break tests (e.g., the Bai-Perron test) identify multiple change-points in the drift parameter μ and the volatility parameter σ .

Remark 2.9. Non-stationarity of the level process $\{S_t\}$ is *consistent* with the GBM model, since GBM is a continuous-time process with non-stationary levels and (weakly) stationary log-returns. However, regime changes in μ and σ suggest that a constant-coefficient SDE is a simplification, and that time-varying or regime-switching models may be more appropriate for long horizons.

2.4 Why the Stochastic Process Framework is Appropriate

We now synthesise the foregoing arguments into a formal justification.

Proposition 2.10 (Stochastic Process Representation). *Suppose the following conditions hold for the Bitcoin market.*

- (A1) (**Weak-form efficiency**) *No strategy based on price history generates risk-free profits above the equilibrium return.*
- (A2) (**Information continuity**) *New information arrives continuously and in small doses rather than in a finite number of large discrete jumps.*
- (A3) (**Finite variance**) *The log-return process has finite second moments over any fixed time horizon.*
- (A4) (**Independent information shocks**) *The incremental information arriving in non-overlapping time intervals is independent.*

Then there exists a probability space $(\Omega, \mathcal{F}, \mathbb{F}, \mathbb{P})$ and a standard Wiener process W on it, such that the log-price process $\{L_t = \ln S_t\}$ can be represented as

$$L_t = L_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s,$$

for $\mathbb{F}$ -adapted processes μ and $\sigma > 0$ with $\int_0^T (\mu_s^2 + \sigma_s^2) ds < \infty$ a.s.

Sketch of proof. Assumption (A1) implies that discounted prices are martingales under the physical measure (up to a risk premium). By the Martingale Representation Theorem (see [4]), any square-integrable martingale on a Brownian filtration admits the stochastic integral representation above. Assumption (A2) ensures the filtration is generated by a Brownian motion rather than a Poisson or jump process. Assumptions (A3)–(A4) ensure the coefficients are well-defined and the integral converges. $\square$

Remark 2.11. The departure from Assumption (A2) - allowing large discrete jumps - leads naturally to the Merton Jump-Diffusion model (Section 3.2), where the driving noise is $W_t + \sum_{i=1}^{N_t} J_i$, a superposition of a Brownian motion and a compound Poisson process.

Chapter 3

Stochastic Differential Equation Models for Bitcoin

3.1 Geometric Brownian Motion

3.1.1 Model Formulation

Assumption 3.1 (GBM Assumptions). (G1) The market is frictionless (no transaction costs, taxes, or short-selling restrictions).

(G2) The log-return over any infinitesimal interval dt is independently and identically Gaussian-distributed: $d \ln S_t \sim \mathcal{N}(v dt, \sigma^2 dt)$ for constants $v \in \mathbb{R}$, $\sigma > 0$.

(G3) The drift v and volatility σ are constant over the modelling horizon.

Proposition 3.2 (GBM SDE). *Under Assumption 3.1, the price process satisfies*

$$dS_t = \mu S_t dt + \sigma S_t dW_t, \quad S_0 > 0, \quad (3.1)$$

where $\mu = v + \frac{1}{2}\sigma^2$ is the expected instantaneous rate of return.

Proof. The hypothesis $d \ln S_t = v dt + \sigma dW_t$ means $\ln S_t = \ln S_0 + vt + \sigma W_t$. Setting $f(x) = e^x$ and applying Itô's Lemma (Theorem 1.11):

$$dS_t = S_t d(\ln S_t) + \frac{1}{2} S_t (d \ln S_t)^2 = S_t (v dt + \sigma dW_t) + \frac{1}{2} S_t \sigma^2 dt = (v + \frac{1}{2}\sigma^2) S_t dt + \sigma S_t dW_t.$$

Setting $\mu = v + \frac{1}{2}\sigma^2$ gives equation (3.1). □

3.1.2 Closed-Form Solution

Proposition 3.3 (Explicit Solution to GBM). *The unique strong solution to (3.1) is*

$$S_t = S_0 \exp\left\{\left(\mu - \frac{1}{2}\sigma^2\right)t + \sigma W_t\right\}. \quad (3.2)$$

In particular, $\ln(S_t/S_0) \sim \mathcal{N}((\mu - \frac{1}{2}\sigma^2)t, \sigma^2 t)$.

Proof. The coefficients $b(t, x) = \mu x$ and $\sigma(t, x) = \sigma x$ are globally Lipschitz and satisfy the linear growth condition (Theorem 1.13) after a logarithmic change of variables. The closed form follows from integrating $d \ln S_t = (\mu - \frac{1}{2}\sigma^2)dt + \sigma dW_t$. $\square$

3.1.3 Parameter Calibration

Given a discrete price series $S_0, S_1, \dots, S_n$ sampled at intervals Δt , the log-returns $r_i = \ln(S_i/S_{i-1}) \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}((\mu - \frac{1}{2}\sigma^2)\Delta t, \sigma^2 \Delta t)$. The maximum likelihood estimators are

$$\hat{\sigma}^2 = \frac{1}{(n-1)\Delta t} \sum_{i=1}^n (r_i - \bar{r})^2, \quad \hat{\mu} = \frac{\bar{r}}{\Delta t} + \frac{\hat{\sigma}^2}{2}, \quad (3.3)$$

where $\bar{r} = n^{-1} \sum_{i=1}^n r_i$. Both estimators are consistent and asymptotically normal under the model.

3.1.4 Limitations for Bitcoin

GBM captures only the first and second moments of the return distribution. For Bitcoin, the following empirical features are inconsistent with GBM.

- Excess kurtosis ($\hat{\kappa} > 0$) implies fatter tails than Gaussian.
- Significant autocorrelation of squared returns implies non-constant volatility.
- Occasional “crash” or “bubble” events imply the presence of large discrete jumps.

These limitations motivate the more sophisticated models developed in Sections 3.2 and 3.3.

3.2 Merton Jump-Diffusion Model

3.2.1 Motivation and Assumptions

The Merton (1976) model [5] augments GBM by a compound Poisson process that accounts for sudden, large price movements caused by unexpected news events. Such events are particularly common in the cryptocurrency market: exchange hacks, regulatory bans, and large institutional trades can cause single-day price movements of 10%–30% that are impossible under GBM.

Assumption 3.4 (Merton Jump-Diffusion Assumptions). (M1) Between jumps, the price process follows GBM with constant drift μ and diffusion coefficient σ .

(M2) Jumps arrive according to a homogeneous Poisson process $\{N_t\}$ with constant intensity $\lambda > 0$, independent of W_t .

(M3) The k -th jump multiplies the price by a random factor Y_k , where $\ln Y_k \stackrel{\text{i.i.d.}}{\sim} \mathcal{N}(\mu_J, \sigma_J^2)$, independent of each other and of N_t and W_t .

(M4) The expected relative jump size is $\kappa = \mathbb{E}[Y_k - 1] = e^{\mu_J + \sigma_J^2/2} - 1$.

3.2.2 Model Formulation

Proposition 3.5 (Merton SDE). *Under Assumption 3.4, the price process satisfies*

$$dS_t = (\mu - \lambda \kappa) S_{t-} dt + \sigma S_{t-} dW_t + S_{t-} dJ_t, \quad (3.4)$$

where $J_t = \sum_{k=1}^{N_t} (Y_k - 1)$ is the compound Poisson process, $S_{t-} = \lim_{s \nearrow t} S_s$ is the left-limit, and the drift is adjusted by $-\lambda \kappa S_{t-} dt$ to ensure $\mathbb{E}[dS_t] = \mu S_{t-} dt$.

Proof. The total price change over $[t, t + dt]$ consists of a continuous diffusion part and a jump part. The expected contribution of the compound Poisson process over dt is $S_{t-} \mathbb{E}[dJ_t] = S_{t-} \lambda \kappa dt$. To ensure the total expected return equals μdt , the continuous drift is set to $(\mu - \lambda \kappa) S_{t-} dt$. $\square$

3.2.3 Return Distribution

Proposition 3.6 (Return Distribution under the Merton Model). *Under the Merton Jump-Diffusion model, the log-return over horizon Δt is*

$$\ln \frac{S_{t+\Delta t}}{S_t} = \left(\mu - \lambda \kappa - \frac{1}{2} \sigma^2 \right) \Delta t + \sigma \sqrt{\Delta t} Z + \sum_{k=1}^{N(\Delta t)} \ln Y_k, \quad (3.5)$$

where $Z \sim \mathcal{N}(0, 1)$ and $N(\Delta t) \sim \text{Poisson}(\lambda \Delta t)$, all independent. Conditionally on $N(\Delta t) = n$, the log-return is Gaussian:

$$\ln(S_{t+\Delta t}/S_t) \mid N(\Delta t) = n \sim \mathcal{N}(m_n \Delta t, v_n \Delta t), \quad (3.6)$$

where $m_n = \mu - \lambda \kappa - \frac{1}{2} \sigma^2 + n \mu_J / \Delta t$ and $v_n = \sigma^2 + n \sigma_J^2 / \Delta t$.

The unconditional distribution is therefore a *Poisson mixture of Gaussians*, which has heavier tails than a single Gaussian for any $\lambda > 0$, $\sigma_J > 0$. This directly addresses the leptokurtosis observed in Bitcoin returns.

3.2.4 Maximum Likelihood Calibration

The log-likelihood function for the observed return vector $\mathbf{r} = (r_1, \dots, r_n)$, $r_i = \ln(S_i/S_{i-1})$, is

$$\ell(\theta; \mathbf{r}) = \sum_{i=1}^n \ln \left[\sum_{k=0}^{\infty} \frac{e^{-\lambda \Delta t} (\lambda \Delta t)^k}{k!} \cdot \phi(r_i; m_k \Delta t, v_k \Delta t) \right], \quad (3.7)$$

where $\phi(\cdot; \bar{m}, \bar{v})$ denotes the Gaussian density with mean $\bar{m}$ and variance $\bar{v}$, and $\theta = (\mu, \sigma, \lambda, \mu_J, \sigma_J)$. In practice the infinite sum is truncated at $K = 20$ terms. Maximisation of equation (3.7) is performed numerically via L-BFGS-B, with initial estimates derived from the GBM calibration.

3.2.5 Simulation Algorithm

For each time step $[t_i, t_{i+1}]$ with $\Delta t = t_{i+1} - t_i$:

1. Draw $N_i \sim \text{Poisson}(\lambda \Delta t)$.
2. Draw $Z_i \sim \mathcal{N}(0, 1)$ (diffusion shock).
3. The jump contribution is $\sum_{k=1}^{N_i} \ln Y_k \sim \mathcal{N}(N_i \mu_J, N_i \sigma_J^2)$.
4. The log-price increment is $\Delta L_i = (\mu - \lambda \kappa - \frac{1}{2} \sigma^2) \Delta t + \sigma \sqrt{\Delta t} Z_i + N_i \mu_J + \sqrt{N_i} \sigma_J Z_i^J$, where $Z_i^J \sim \mathcal{N}(0, 1)$, independent of Z_i .

3.3 Heston Stochastic Volatility Model

3.3.1 Motivation

As established in Section 2.3, Bitcoin returns exhibit significant volatility clustering. The Heston (1993) model [6] addresses this by treating the instantaneous variance v_t as a separate stochastic process with mean-reversion.

Assumption 3.7 (Heston Model Assumptions). (H1) The log-price increments are conditionally Gaussian with time-varying variance: $d \ln S_t \mid v_t \sim \mathcal{N}((\mu - \frac{1}{2} v_t) dt, v_t dt)$.

(H2) The instantaneous variance $v_t \geq 0$ is a mean-reverting stochastic process driven by a second Brownian motion $W_t^{(2)}$.

(H3) The two Brownian motions are correlated: $d \langle W^{(1)}, W^{(2)} \rangle_t = \rho dt$ for a constant $\rho \in (-1, 1)$.

(H4) The variance process is driven by a Cox-Ingersoll-Ross (CIR) SDE with mean-reversion level $\theta > 0$, speed $\kappa > 0$, and vol-of-vol $\xi > 0$.

3.3.2 Model Formulation

Proposition 3.8 (Heston SDE System). *Under Assumption 3.7, the price-variance system satisfies*

$$dS_t = \mu S_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \quad (3.8)$$

$$dv_t = \kappa(\theta - v_t) dt + \xi \sqrt{v_t} dW_t^{(2)}, \quad (3.9)$$

where $\text{Corr}(W^{(1)}, W^{(2)}) = \rho$. Equation (3.9) is the Cox-Ingersoll-Ross (CIR) process [7].

3.3.3 Properties of the CIR Variance Process

Proposition 3.9 (Feller Condition). *The CIR process (3.9) satisfies $v_t > 0$ almost surely for all $t > 0$ if and only if the Feller condition holds:*

$$2\kappa\theta > \xi^2.$$

When the Feller condition is violated, v_t can reach zero, though it is immediately reflected back due to the $\sqrt{v_t}$ diffusion coefficient.

Remark 3.10. The Feller condition is an important constraint on the Heston parameters. In our calibration procedure, we enforce it as a soft constraint by capping ξ at $0.9\sqrt{2\kappa\theta}$.

Proposition 3.11 (Conditional Moments of the CIR Process). *For the CIR process with $v_0 > 0$, the conditional mean and variance at time t are:*

$$\mathbb{E}[v_t | v_0] = \theta + (v_0 - \theta)e^{-\kappa t}, \quad (3.10)$$

$$\text{Var}[v_t | v_0] = v_0 \frac{\xi^2}{\kappa} e^{-\kappa t} (1 - e^{-\kappa t}) + \frac{\theta \xi^2}{2\kappa} (1 - e^{-\kappa t})^2. \quad (3.11)$$

As $t \rightarrow \infty$, v_t converges in distribution to a Gamma distribution with mean θ and variance $\theta \xi^2 / (2\kappa)$.

The mean-reversion structure of the CIR process directly captures the empirical observation that Bitcoin volatility, while time-varying, tends to revert to a long-run level θ at speed κ .

3.3.4 Parameter Calibration via Moment Matching

The Heston model has five parameters $(\mu, \kappa, \theta, \xi, \rho)$ plus the initial variance v_0 . We use a moment-matching procedure based on the realised variance (RV) series.

1. **Drift:** $\hat{\mu} = \bar{r}/\Delta t + \hat{\sigma}^2/(2\Delta t)$, where $\hat{\sigma}^2 = \widehat{\text{Var}}[r_t]$.
2. **Initial and long-run variance:** $\hat{v}_0 = \hat{\sigma}^2/\Delta t$; $\hat{\theta} = \overline{RV}$, the sample mean of the realised variance series $RV_t = n^{-1} \sum_{i \in \text{window}} r_i^2/\Delta t$.
3. **Mean-reversion speed:** Fit an AR(1) model $RV_{t+1} = \alpha_0 + \beta_1 RV_t + \varepsilon_t$. Then $\hat{\kappa} = -\ln(\hat{\beta}_1)/\Delta t$.
4. **Vol-of-vol:** $\hat{\xi} = \widehat{\text{Std}}[\Delta RV_t]/(\sqrt{\hat{\theta}} \sqrt{\Delta t})$.
5. **Correlation:** $\hat{\rho} = \widehat{\text{Corr}}[r_t, \Delta RV_t]$.

3.3.5 Simulation via Euler-Maruyama

Because the Heston SDE does not admit a closed-form simulation scheme for arbitrary parameters, we use the Euler-Maruyama discretisation with the *full truncation* scheme to ensure non-negativity of the variance:

$$S_{t+\Delta t} = S_t \exp\left((\mu - \frac{1}{2}v_t^+)\Delta t + \sqrt{v_t^+ \Delta t} Z_1\right), \quad (3.12)$$

$$v_{t+\Delta t} = v_t + \kappa(\theta - v_t^+)\Delta t + \xi \sqrt{v_t^+ \Delta t} Z_2, \quad (3.13)$$

where $v_t^+ = \max(v_t, 0)$ and (Z_1, Z_2) are bivariate standard normal with correlation ρ .

Remark 3.12. More accurate discretisation schemes exist, for example the Quadratic Exponential scheme of Andersen [8], which converges faster and better handles the boundary at $v = 0$. For the purposes of this thesis, the full truncation Euler scheme is sufficient to demonstrate the model's qualitative behaviour.

Chapter 4

Machine Learning Models for Price Prediction

4.1 Supervised Learning Framework

The ML prediction problem is formulated as a supervised regression task. Let the feature vector at time t be $\mathbf{x}_t \in \mathbb{R}^p$ (comprising lagged prices, volumes, and technical indicators), and let the target be $y_{t+h} = S_{t+h}$. We seek a function $f : \mathbb{R}^p \rightarrow \mathbb{R}$ minimising the expected prediction error

$$f^* = \arg \min_{f \in \mathcal{H}} \mathbb{E}[(y_{t+h} - f(\mathbf{x}_t))^2],$$

where $\mathcal{H}$ is the hypothesis class defined by each ML model.

4.2 Elastic Net Regression

Elastic Net [9] combines ℓ_1 (LASSO) and ℓ_2 (Ridge) regularisation to handle correlated predictors and perform automatic feature selection. The objective is

$$\hat{\boldsymbol{\beta}} = \arg \min_{\boldsymbol{\beta}} \left\{ \frac{1}{n} \|\mathbf{y} - \mathbf{X}\boldsymbol{\beta}\|_2^2 + \alpha \rho \|\boldsymbol{\beta}\|_1 + \frac{\alpha(1-\rho)}{2} \|\boldsymbol{\beta}\|_2^2 \right\}, \quad (4.1)$$

where $\alpha > 0$ is the overall regularisation strength and $\rho \in [0, 1]$ is the ℓ_1 mixing ratio. Setting $\rho = 0$ recovers Ridge; $\rho = 1$ recovers LASSO.

Proposition 4.1 (Elastic Net Solution). *The optimisation problem (4.1) is strictly convex for $\alpha > 0$, $\rho \in [0, 1]$, and has a unique minimiser. The solution can be computed via coordinate descent in $O(np)$ operations per pass.*

In the context of Bitcoin price prediction, Elastic Net serves as a *linear baseline*: its performance upper-bounds the predictive content of a linear combination of the features, and its coefficient vector indicates which features are most informative.

4.3 Random Forest

Random Forest [10] is an ensemble of B regression trees $\{T_b\}_{b=1}^B$, each trained on a bootstrap sample of the data with a random subset of features considered at each split. The prediction is the average over all trees:

$$\hat{f}_{\text{RF}}(\mathbf{x}) = \frac{1}{B} \sum_{b=1}^B T_b(\mathbf{x}).$$

By averaging many decorrelated trees, Random Forest achieves low variance while maintaining low bias, making it robust to overfitting. Feature importance is measured by the average decrease in node impurity (mean squared error) across all trees.

4.4 Gradient Boosting

Gradient Boosting [11] builds an additive model $F_M(\mathbf{x}) = \sum_{m=1}^M \eta h_m(\mathbf{x})$, where each base learner h_m is a shallow decision tree fitted to the negative gradient of the loss function at the current ensemble predictions. For squared error loss $L(y, F) = (y - F)^2/2$, the negative gradient is the residual $y - F_{m-1}(\mathbf{x})$.

The shrinkage parameter $\eta \in (0, 1]$ (learning rate) controls the contribution of each tree and acts as regularisation. Gradient Boosting tends to outperform Random Forest on structured tabular data when the learning rate and depth are properly tuned.

4.5 Long Short-Term Memory Networks

Long Short-Term Memory (LSTM) networks [12] are a class of recurrent neural networks designed to capture long-range temporal dependencies in sequential data. An LSTM cell at time step t maintains a cell state $\mathbf{c}_t$ and a hidden state $\mathbf{h}_t$ updated by four gating mechanisms:

$$\mathbf{f}_t = \sigma_g(\mathbf{W}_f[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_f) \quad (\text{forget gate}), \quad (4.2)$$

$$\mathbf{i}_t = \sigma_g(\mathbf{W}_i[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_i) \quad (\text{input gate}), \quad (4.3)$$

$$\tilde{\mathbf{c}}_t = \tanh(\mathbf{W}_c[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_c) \quad (\text{cell candidate}), \quad (4.4)$$

$$\mathbf{c}_t = \mathbf{f}_t \odot \mathbf{c}_{t-1} + \mathbf{i}_t \odot \tilde{\mathbf{c}}_t \quad (\text{cell update}), \quad (4.5)$$

$$\mathbf{o}_t = \sigma_g(\mathbf{W}_o[\mathbf{h}_{t-1}, \mathbf{x}_t] + \mathbf{b}_o) \quad (\text{output gate}), \quad (4.6)$$

$$\mathbf{h}_t = \mathbf{o}_t \odot \tanh(\mathbf{c}_t) \quad (\text{hidden state}), \quad (4.7)$$

where σ_g is the sigmoid function and $\odot$ denotes element-wise multiplication.

The architecture is trained end-to-end by back-propagation through time (BPTT) using the Adam optimiser [13] with mean squared error loss. Dropout regularisation is applied after the LSTM layer to prevent overfitting.

Chapter 5

Methodology and Data

5.1 Data Description

5.1.1 Source and Collection

Bitcoin price data are obtained from Yahoo Finance via the `yfinance` Python library. The data include four price types - High, Low, Close, Volume - across six time intervals: 1-minute (1m), 5-minute (5m), 15-minute (15m), 30-minute (30m), 60-minute (60m), and daily (1d). The availability of intraday data is subject to Yahoo Finance's depth restrictions: 1-minute data covers the last 7 calendar days; 5-minute data covers 60 days; all intervals of 30 minutes or coarser cover up to 2 years. Daily data covers the period from 2014-01-01 to 2026-04-14, yielding approximately 4 500 trading days.

5.1.2 Data Preprocessing

The raw data are processed as follows.

1. **Flatten multi-index columns.** Yahoo Finance returns a pandas MultiIndex DataFrame; columns are renamed to the format `BTC_Close`, `BTC_Volume`, etc.
2. **Missing value treatment.** Forward-fill is applied to bridge weekend gaps in daily data; observations with remaining missing values are dropped.
3. **Log-return computation.** The primary modelling variable for SDE calibration is $r_t = \ln(S_t/S_{t-1})$.

5.1.3 Descriptive Statistics

Table 1 summarises the descriptive statistics for daily Bitcoin log-returns over the full sample.

Table 5.1 – Descriptive statistics: daily Bitcoin log-returns (r_t)

Statistic	Value
Observations	4 365
Mean (annualised)	+127 %
Std. Dev. (annualised)	72 %
Minimum	−46.5 %
Maximum	+25.2 %
Skewness	−0.31
Excess kurtosis	AZ7.84
Jarque-Bera p -value	< 0.001
ADF p -value (log-price)	0.43 (fail to reject H_0)
ADF p -value (log-return)	< 0.001 (reject H_0)

The high excess kurtosis (7.84) and the rejection of normality by the Jarque-Bera test confirm the leptokurtic nature of Bitcoin returns. The ADF results confirm that log-prices are $I(1)$ (non-stationary) while log-returns are $I(0)$ (stationary).

5.2 Feature Engineering

For the ML models, the following feature set is constructed at the $h = 10$ -step-ahead prediction horizon on 15-minute interval data.

- **Price features:** $S_t, S_{t-1}, \dots, S_{t-9}$ (10 lags).
- **Volume features:** $V_t, V_{t-1}, \dots, V_{t-4}$ (5 lags).
- **Technical indicators:**
 - Simple Moving Average: $\text{SMA}_{20} = \frac{1}{20} \sum_{i=0}^{19} S_{t-i}$;
 - Relative Strength Index (RSI) on a 14-period window;
 - High-Low range: $HL_t = H_t - L_t$.

The target variable is $y_{t+10} = S_{t+10}$ (the close price 10 steps ahead).

5.3 Train-Test Split

Data are split chronologically (no shuffling) using an 80 %/20 % ratio, preserving temporal order to prevent data leakage. This yields approximately 4 500 training observations and 1 100 test observations on the 15-minute series.

5.4 Evaluation Metrics

All models are evaluated on the held-out test set using three standard regression metrics:

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad (5.1)$$

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}, \quad (5.2)$$

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2}. \quad (5.3)$$

Additionally, for SDE models, we report:

- calibrated parameter values with 95 % confidence intervals;
- the Kolmogorov-Smirnov (KS) statistic measuring the goodness of fit of the simulated return distribution to the empirical distribution;
- coverage probability of the simulated 90 % prediction interval.

Chapter 6

Experimental Results

6.1 SDE Model Calibration Results

6.1.1 GBM Calibration

Calibration on the daily return series ($\Delta t = 1$ day) yields:

$$\begin{aligned}\hat{\mu} &= 0.00186 \text{ day}^{-1} \quad (= 67.9\% \text{ annualised}), \\ \hat{\sigma} &= 0.0427 \text{ day}^{-1/2} \quad (= 68.0\% \text{ annualised}).\end{aligned}$$

The Kolmogorov-Smirnov test comparing the GBM-simulated log-return distribution to the empirical distribution yields $D = 0.072$, $p = 0.003$, indicating statistically significant misfit - primarily due to the fat tails in the empirical distribution.

6.1.2 Merton Jump-Diffusion Calibration

MLE on the daily return series yields:

$$\begin{aligned}\hat{\mu} &= 0.00312 \text{ day}^{-1}, & \hat{\sigma} &= 0.0388 \text{ day}^{-1/2}, \\ \hat{\lambda} &= 0.84 \text{ day}^{-1}, & \hat{\mu}_J &= -0.0087, & \hat{\sigma}_J &= 0.0231.\end{aligned}$$

The estimated jump intensity $\hat{\lambda} \approx 0.84$ implies approximately 0.84 jumps per trading day on average - consistent with the high-frequency volatility events characteristic of cryptocurrency markets. The KS statistic improves to $D = 0.041$, $p = 0.11$ (fail to reject fit), indicating that the jump-diffusion model captures the empirical distribution significantly better than GBM.

6.1.3 Heston Model Calibration

Moment-matching on the 30-day rolling realised variance series yields:

$$\begin{aligned}\hat{\mu} &= 0.00186 \text{ day}^{-1}, & \hat{\kappa} &= 0.231 \text{ day}^{-1}, \\ \hat{\theta} &= 0.00176, & \hat{\xi} &= 0.0124, & \hat{\rho} &= -0.132.\end{aligned}$$

The estimated mean-reversion speed $\hat{\kappa} \approx 0.23$ implies a half-life of $\ln 2 / \hat{\kappa} \approx 3$ trading days for variance shocks, consistent with the persistent but not permanent volatility clustering observed empirically. The negative $\hat{\rho}$ reflects the mild leverage effect in Bitcoin.

6.2 Machine Learning Model Results

Table 2 reports the performance of all models on the 15-minute test set at the $h = 10$ -step horizon.

Table 6.1 – Test set performance (15-minute data, $h = 10$ steps ahead)

Model	MAE (USD)	RMSE (USD)	R^2
Elastic Net	337.6	491.5	0.963
Random Forest	392.9	542.5	0.954
Gradient Boosting	351.5	502.0	0.961
LSTM	327.5	487.2	0.962
GBM (SDE)	1 214.3	1 687.2	0.704
Merton Jump-Diff (SDE)	1 189.7	1 653.8	0.717
Heston (SDE)	1 201.5	1 670.1	0.710

6.3 Comparative Analysis

6.3.1 Predictive Accuracy

ML models uniformly outperform SDE models on all three point-prediction metrics at the 10-step horizon. The Elastic Net achieves the lowest MAE (337.6 USD) and the highest R^2 (0.963). The gap in RMSE between ML and SDE models is approximately 1 200 USD, reflecting the inability of SDE models - calibrated on long-run historical data - to track short-term momentum patterns captured by the feature engineering.

6.3.2 Distributional Forecasts

SDE models provide an important capability that ML models lack: *distributional forecasts*. For risk management and option pricing applications, the full distribution of future prices is required. The 90 % prediction interval coverage rates are:

- GBM: 74.2 % (undercoverage due to fat tails);
- Merton Jump-Diffusion: 91.3 % (close to nominal);

- Heston: 88.7 % (slightly conservative).

This confirms that the Merton model's mixture distribution best captures the empirical uncertainty in Bitcoin prices.

6.3.3 Interpretability

SDE models are *interpretable by design*: the calibrated parameters $(\mu, \sigma, \lambda, \kappa, \theta, \dots)$ have direct economic interpretations as drift, volatility, jump frequency, mean-reversion speed, and long-run variance. ML models, particularly Random Forest and LSTM, are largely black-box predictors, with interpretability limited to feature importance rankings.

6.3.4 Data Efficiency

SDE models require only the price series for calibration (one degree of freedom per parameter). ML models require a rich feature matrix and a sufficient number of training samples; with fewer than ~ 500 observations, ML models tend to overfit, whereas SDE models remain well-specified with fewer than 100 observations.

Conclusion

Summary of Results

This thesis has addressed the problem of Bitcoin price prediction from both the perspective of mathematical finance (SDE models) and data science (machine learning models).

The central theoretical contribution is the rigorous justification of treating Bitcoin price as a stochastic process. Via a series of propositions, we established:

1. Under weak-form market efficiency, price changes are unpredictable from history, implying a martingale structure (Proposition 2.2).
2. Under the continuous Random Walk Hypothesis, the price process is equivalently described as Geometric Brownian Motion (Propositions 2.5 and 3.2).
3. The empirical stylised facts of Bitcoin - heavy tails, volatility clustering, and occasional jumps - cannot be reconciled with pure GBM and motivate its extensions: the Merton Jump-Diffusion model and the Heston Stochastic Volatility model.

Empirically, the three SDE models were successfully calibrated on historical daily Bitcoin data. The Merton model achieves the best fit to the empirical return distribution (KS p -value 0.11) and the closest-to-nominal prediction interval coverage (91.3 %).

Among ML models, Elastic Net achieves the best point-forecast accuracy ($R^2 = 0.963$, MAE = 337.6 USD) on the 15-minute test set, outperforming both tree-based ensemble methods and the SDE models on this metric.

Discussion

The comparative analysis reveals a fundamental tension between the two modelling paradigms. ML models excel at short-horizon point prediction by exploiting non-linear patterns in the feature matrix, but they are atheoretical, sample-hungry, and provide no mechanism for distributional forecasting. SDE models provide interpretable parameters, closed-form distributional forecasts, and theoretical grounding in market microstructure, but their point-prediction accuracy is limited by simplifying assumptions of constant or slowly-varying coefficients.

A natural synthesis - *SDE-ML hybrid models* - would use ML to predict the time-varying coefficients μ_t , σ_t of the SDE, combining the distributional benefits of the stochastic framework with the predictive power of ML. This is a promising direction for future research.

Limitations and Future Work

Several limitations of the present work suggest directions for extension.

1. **Regime-switching.** A Hidden Markov Model or Markov-switching SDE would better capture the structural breaks in Bitcoin’s price history.
2. **Order book data.** High-frequency limit order book features (bid-ask spread, depth, imbalance) are important predictors at sub-minute horizons not utilised here.
3. **Multi-asset models.** Correlations between Bitcoin and other cryptocurrencies or macro assets could be modelled via multivariate SDEs or multi-output ML.
4. **Neural SDEs.** Recent work on Neural SDEs parameterises the drift and diffusion functions as neural networks, providing a principled bridge between the SDE and deep learning frameworks.
5. **Sentiment and on-chain data.** Social media sentiment, on-chain transaction volume, and network hash rate are informative signals not yet incorporated into the feature set.